

Olympiad Test Paper — Mathematics (Class 7) — Paper 4

Chapter 11: Finding Common Ground

Paper 4 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	11 — Finding Common Ground
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Decide HCF (largest divider — share/cut/tile) vs LCM (smallest common multiple — meet/ring again); take LOWEST powers of common primes for HCF and HIGHEST powers of all primes for LCM; for 'same unknown remainder' use HCF of pairwise differences, for 'leaves remainder r' use $\text{LCM} \cdot k + r$, and remember $\text{HCF} \cdot \text{LCM} = \text{product}$ only for two numbers while $\text{LCM} = \text{product}$ only for co-primes. Do not write on this paper — mark answers on your answer sheet.

1. The numbers 1764 and 2940 factorise as $1764 = 2^2 \times 3^2 \times 7^2$ and $2940 = 2^2 \times 3 \times 5 \times 7^2$. Their HCF is:

(A) $2^2 \times 3^2 \times 5 \times 7^2 = 8820$

(B) $2^2 \times 7^2 = 196$

(C) $2^2 \times 3 \times 7^2 = 588$

(D) $2^2 \times 3 \times 7 = 84$

2. For $1764 = 2^2 \times 3^2 \times 7^2$ and $2940 = 2^2 \times 3 \times 5 \times 7^2$, the LCM is:

(A) $2^2 \times 3 \times 7^2 = 588$

(B) $2^2 \times 3^2 \times 5^2 \times 7^2 = 44100$

(C) $2^4 \times 3^3 \times 5 \times 7^4$

(D) $2^2 \times 3^2 \times 5 \times 7^2 = 8820$

3. Two numbers are $2^4 \times 3^2 \times 5$ and $2^3 \times 3^3 \times 5^2$. The ratio LCM : HCF equals:

(A) 6

(B) 150

(C) 30

(D) 5

4. How many factors of 720 are perfect squares (including 1)?

(A) 4

(B) 6

(C) 8

(D) 3

5. The total number of factors of 5400 is:

- (A) 24 (B) 40
(C) 18 (D) 48

6. How many factors of 600 are even?

- (A) 24 (B) 12
(C) 18 (D) 6

7. The smallest natural number that has exactly 12 factors AND is a multiple of 12 is:

- (A) 96 (B) 60
(C) 72 (D) 36

8. A number $N = 2^a \times 3^b$ has exactly 15 factors, with $a > b \geq 1$. The value of N is:

- (A) 324 (B) 48
(C) 144 (D) 162

9. The largest 3-digit number of the form $2^a \times 3^b$ (with $a \geq 1$ and $b \geq 1$) is:

- (A) 864 (B) 768
(C) 648 (D) 972

10. Two numbers are 540 and 1260. Their HCF is:

- (A) 60 (B) 3780
(C) 90 (D) 180

11. The LCM of 540 and 1260 is:

- (A) 180 (B) 3780
(C) 11340 (D) 1890

12. The number of common factors of 360 and 540 is:

- (A) 12 (B) 9
(C) 24 (D) 18

13. The product of two numbers is 7560 and their HCF is 6. Their LCM is:

- (A) 1260 (B) 45360
(C) 1266 (D) 210

14. Two numbers have HCF 14 and LCM 1764. If one number is 98, the other is:

- (A) 126 (B) 1764
(C) 252 (D) 12

15. The product of two numbers equals 18 times the square of their HCF. If their HCF is 12, their LCM is:

- (A) 216 (B) 144
(C) 2592 (D) 18

16. For two numbers, $\text{LCM} = 14 \times \text{HCF}$ and the HCF is 15. The product of the two numbers is:

- (A) 3150 (B) 210
(C) 225 (D) 31500

17. Two numbers are in the ratio 8 : 15 and their HCF is 9. Their LCM is:

- (A) 120 (B) 135
(C) 1080 (D) 8640

18. Two numbers in the ratio 5 : 8 have an LCM of 360. Their HCF is:

- (A) 40 (B) 45
(C) 9 (D) 72

19. The HCF of two numbers is 12 and their LCM is 720. The number of unordered pairs (a, b) with $a < b$ is:

- (A) 2 (B) 4
(C) 3 (D) 1

20. The HCF of two numbers is 15 and their LCM is 3150. How many unordered pairs (a, b) with $a < b$ are possible?

- (A) 4 (B) 3
(C) 16 (D) 8

21. Two numbers have HCF 7 and their LCM is $7 \times 11 \times 13 \times 17$. How many unordered pairs (a, b) with $a < b$ exist?

- (A) 8 (B) 2
(C) 4 (D) 3

22. The sum of two numbers is 384 and their HCF is 48. How many unordered pairs of such numbers are possible?

- (A) 2 (B) 4
(C) 1 (D) 3

23. A number n satisfies $\text{HCF}(n, 36) = 12$ and $\text{LCM}(n, 36) = 360$. The number n is:

- (A) 60 (B) 120
(C) 24 (D) 48

24. Which of these CANNOT be the HCF and LCM of some pair of numbers?

- (A) HCF 12, LCM 144 (B) HCF 9, LCM 270
(C) HCF 15, LCM 300 (D) HCF 14, LCM 204

25. A florist has 252 roses, 378 lilies and 630 tulips. She makes identical bouquets using all the flowers, each bouquet having the same number of each kind. The largest number of bouquets is:

- (A) 63 (B) 18
(C) 126 (D) 42

26. For the florist in the previous setting (252 roses, 378 lilies, 630 tulips, 126 bouquets), the number of roses in each bouquet is:

- | | |
|-------|-------|
| (A) 3 | (B) 2 |
| (C) 5 | (D) 6 |

27. Square tiles of the largest possible whole-cm side exactly pave a room 7 m 92 cm long and 5 m 40 cm wide. The side of each tile is:

- | | |
|------------|-----------|
| (A) 12 cm | (B) 18 cm |
| (C) 108 cm | (D) 36 cm |

28. For the room of the previous setting (792 cm by 540 cm, tile side 36 cm), the least number of square tiles required is:

- | | |
|---------|---------|
| (A) 132 | (B) 330 |
| (C) 165 | (D) 300 |

29. Three ropes of lengths 9 m, 13.5 m and 18.9 m are cut into equal pieces of the greatest possible length with nothing wasted. That length is:

- | | |
|------------|-----------|
| (A) 90 cm | (B) 45 cm |
| (C) 270 cm | (D) 30 cm |

30. A merchant has 437 L, 551 L and 703 L of three oils. The largest can that measures each an exact whole number of times has capacity:

- | | |
|----------|----------|
| (A) 23 L | (B) 17 L |
| (C) 19 L | (D) 29 L |

31. A gardener has 1080 marigolds and 1320 roses, arranging each kind into rows of equal, largest possible size (one kind per row). The number of flowers per row is:

- | | |
|---------|---------|
| (A) 60 | (B) 240 |
| (C) 120 | (D) 24 |

32. For the gardener in the previous setting (1080 marigolds, 1320 roses, 120 per row), the total number of rows is:

- | | |
|--------|--------|
| (A) 9 | (B) 20 |
| (C) 11 | (D) 22 |

33. Beads numbering 391 red, 425 blue and 493 green are strung so each string holds one colour only and every string has the same largest number of beads. That number is:

- | | |
|--------|--------|
| (A) 23 | (B) 25 |
| (C) 17 | (D) 29 |

34. A field 1517 m by 1591 m is divided into identical square plots of the greatest possible size. The number of plots along the 1591 m side is:

- | | |
|--------|--------|
| (A) 41 | (B) 37 |
| (C) 43 | (D) 59 |

35. Coloured cards numbering 1224, 1632 and 2040 are bundled so each bundle holds cards of one colour only and all bundles are equal and as large as possible. Each bundle contains:

- | | |
|---------|---------|
| (A) 408 | (B) 204 |
| (C) 136 | (D) 612 |

36. A shopkeeper has 84 kg of rice, 126 kg of wheat and 210 kg of pulses, packing each grain into bags of equal, largest possible size (one grain per bag). The total number of bags is:

- | | |
|--------|--------|
| (A) 10 | (B) 12 |
| (C) 42 | (D) 20 |

37. Three bells ring at intervals of 18, 24 and 32 minutes. Ringing together now, they next ring together after:

- | | |
|-----------------|-----------------|
| (A) 288 minutes | (B) 144 minutes |
| (C) 576 minutes | (D) 96 minutes |

38. Four lights flash every 12, 18, 21 and 28 seconds. After flashing together, the next simultaneous flash is after:

- | | |
|-----------------|-----------------|
| (A) 504 seconds | (B) 126 seconds |
| (C) 84 seconds | (D) 252 seconds |

39. Three signals blink every 12, 21 and 28 minutes. They blink together at 9:00 am. The next time all three blink together is:

- | | |
|--------------|--------------|
| (A) 10:24 am | (B) 11:48 am |
| (C) 9:84 am | (D) 10:00 am |

40. Three runners complete a lap in 56, 70 and 84 seconds. Starting together, the number of complete laps the FASTEST runner finishes by the time they next all meet is:

- | | |
|--------|--------|
| (A) 12 | (B) 10 |
| (C) 14 | (D) 15 |

41. Cyclists lap a track every 54 s, 72 s and 90 s. Starting together, the number of complete laps the slowest cyclist has finished when all three next meet is:

- | | |
|--------|--------|
| (A) 15 | (B) 12 |
| (C) 20 | (D) 10 |

42. Two gears mesh, one with 36 teeth and the other with 48 teeth. After how many teeth of the SMALLER gear pass do they return to their starting alignment?

- | | |
|---------------|----------------|
| (A) 144 teeth | (B) 288 teeth |
| (C) 12 teeth | (D) 1728 teeth |

43. Bells P, Q, R toll every 9, 12 and 15 seconds. In 30 minutes, how many times do all three toll together, NOT counting the toll at the very start?

- (A) 10 (B) 11
(C) 5 (D) 9

44. Four bells toll at intervals of 6, 8, 12 and 16 seconds, starting together. In 4 minutes, how many times do all four toll together INCLUDING the toll at the start?

- (A) 5 (B) 6
(C) 10 (D) 4

45. Two buses leave a stop every 36 minutes and every 48 minutes, together at 7:00 am. They next leave together at:

- (A) 8:24 am (B) 11:48 am
(C) 9:24 am (D) 8:00 am

46. Three traffic lights change every 48 s, 72 s and 108 s. Having just changed together, after how long do all three next change together?

- (A) 432 seconds (B) 216 seconds
(C) 864 seconds (D) 144 seconds

47. The smallest number which, when divided by 15, 20 and 35, leaves a remainder of 8 in each case is:

- (A) 420 (B) 428
(C) 412 (D) 848

48. The largest number that divides 1305, 4665 and 6905 leaving the same remainder in each case is:

- (A) 140 (B) 1120
(C) 560 (D) 2240

49. A number leaves remainder 5 when divided by each of 8, 12 and 18, and is the smallest such number greater than 5. It is:

- (A) 72 (B) 67
(C) 149 (D) 77

50. The greatest 3-digit number that leaves remainder 7 when divided by 8, 12 and 20 is:

- (A) 967 (B) 960
(C) 847 (D) 975

51. The smallest number which, when increased by 11, becomes exactly divisible by 18, 24 and 30 is:

- (A) 360 (B) 371
(C) 709 (D) 349

52. A number leaves remainder 1 when divided by 2, 3, 4, 5 and 6, and remainder 0 when divided by 7. The smallest such number greater than 1 is:

- (A) 61 (B) 121
(C) 301 (D) 421

53. A number when divided by 12 leaves remainder 7, and when divided by 18 leaves remainder 13. The smallest such positive number is:

- (A) 43 (B) 5
(C) 36 (D) 31

54. Which set of three numbers is pairwise co-prime (every pair has HCF 1)?

- (A) 6, 25 and 14 (B) 9, 16 and 35
(C) 8, 9 and 21 (D) 15, 22 and 33

55. If a number is divisible by both 16 and 36, the smallest positive value it can take is:

- (A) 576 (B) 144
(C) 4 (D) 288

56. For two numbers a and b with $\text{HCF}(a, b) = 1$, the value of $\text{HCF}(a^3, b^3)$ is:

- (A) 1 (B) a^3
(C) ab (D) $(ab)^3$

57. Two co-prime numbers, each greater than 1, have a product of 1517. The two numbers are:

- (A) 37 and 41 (B) 1 and 1517
(C) 23 and 66 (D) 31 and 49

58. A number divisible by 9 and by 10 but NOT necessarily by 4 is always divisible by:

- (A) 90 (B) 360
(C) 180 (D) 45

59. Among the numbers from 1 to 500, how many are common multiples of 12 and 15?

- (A) 10 (B) 33
(C) 4 (D) 8

60. The greatest length of a measuring rod that can measure exactly each of 4 m 25 cm, 6 m 80 cm and 10 m 20 cm is:

- (A) 17 cm (B) 85 cm
(C) 5 cm (D) 170 cm

Solutions — Mathematics (Class 7), Chapter 11: Finding Common Ground — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	C	31	C	41	B	51	D
2	D	12	D	22	A	32	B	42	A	52	C
3	C	13	A	23	B	33	C	43	A	53	D
4	B	14	C	24	D	34	C	44	B	54	B
5	D	15	A	25	C	35	A	45	C	55	B
6	C	16	A	26	B	36	D	46	A	56	A
7	B	17	C	27	D	37	A	47	B	57	A
8	C	18	C	28	B	38	D	48	B	58	A
9	D	19	B	29	A	39	A	49	D	59	D
10	D	20	D	30	C	40	D	50	A	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) HCF uses the lowest power of each COMMON prime. Common primes: 2 (min of $2^2, 2^2$) $\rightarrow 2^2$; 3 (min of $3^2, 3^1$) $\rightarrow 3$; 7 (min of $7^2, 7^2$) $\rightarrow 7^2$. The prime 5 is in only one number, so it is excluded. $\text{HCF} = 4 \times 3 \times 49 = 588$. Taking highest powers and including 5 gives the LCM 8820; dropping the common 3 gives 196; lowering the 7 to 7^1 gives 84.

2. (D) LCM uses the highest power of EVERY prime present: 2^2 (both) $\rightarrow 2^2$; 3 (max of $3^2, 3^1$) $\rightarrow 3^2$; 5 (only in 2940) $\rightarrow 5$; 7 (both 7^2) $\rightarrow 7^2$. $\text{LCM} = 4 \times 9 \times 5 \times 49 = 8820$. Using lowest powers gives the HCF 588; raising 5 to 5^2 invents a power not present; adding exponents instead of taking the maximum gives the bloated last form.

3. (C) $\text{HCF (lowest powers)} = 2^3 \times 3^2 \times 5$; $\text{LCM (highest powers)} = 2^4 \times 3^3 \times 5^2$. The ratio $\text{LCM:HCF} = 2^{(4-3)} \times 3^{(3-2)} \times 5^{(2-1)} = 2 \times 3 \times 5 = 30$. Forgetting the 5 gives 6; squaring the 5 gives 150; using only the 5 gives 5.

4. (B) $720 = 2^4 \times 3^2 \times 5$. A perfect-square factor needs even exponents: for 2^4 choose 0, 2 or 4 (3 ways); for 3^2 choose 0 or 2 (2 ways); for 5^1 only 0 (1 way). Total = $3 \times 2 \times 1 = 6$

(they are 1, 4, 16, 9, 36, 144). Using $2 \times 2 = 4$ forgets a choice; allowing 5 gives 8; counting only 1, 4, 9 gives 3.

5. (D) $5400 = 2^3 \times 3^3 \times 5^2$. Number of factors = $(3+1)(3+1)(2+1) = 4 \times 4 \times 3 = 48$. Mis-adding ($4 \times 3 \times 2 = 24$) or using the exponents themselves ($3 \times 3 \times 2 = 18$) gives the traps.

6. (C) $600 = 2^3 \times 3 \times 5^2$. Total factors = $(3+1)(1+1)(2+1) = 24$. The ODD factors use 2^0 only: $1 \times 2 \times 3 = 6$. Even factors = $24 - 6 = 18$. Forgetting to subtract the odd factors gives 24; counting only the odd ones gives 6.

7. (B) A multiple of 12 contains at least $2^2 \times 3$. Try $2^2 \times 3 \times 5 = 60$: factor count $(2+1)(1+1)(1+1) = 12$ ✓. Smaller multiples of 12 — 12, 24, 36, 48 — have 6, 8, 9, 10 factors, none reaching 12. So 60 is smallest. $72 = 2^3 \times 3^2$ also has 12 factors but is larger; $96 = 2^5 \times 3$ has 12 factors but is larger; 36 has only 9 factors.

8. (C) 15 factors means $(a+1)(b+1) = 15 = 5 \times 3$. With $a > b$ take $a+1 = 5$, $b+1 = 3$, so $a = 4$, $b = 2$: $N = 2^4 \times 3^2 = 16 \times 9 = 144$. Swapping to $a = 2$, $b = 4$ (which breaks $a > b$) gives $2^2 \times 3^4 = 324$; using $15 = 15 \times 1$ forces a single prime, impossible here.

9. (D) $972 = 2^2 \times 3^5 = 4 \times 243 = 972 \leq 999$ ✓. Nearby candidates: $2^7 \times 3^2 = 1152$ and $2^4 \times 3^4 = 1296$ and $2 \times 3^6 = 1458$ all exceed 999; $2^5 \times 3^3 = 864$ and $2^8 \times 3 = 768$ and $2^3 \times 3^4 = 648$ are all smaller than 972. Hence 972 is the largest.

10. (D) $540 = 2^2 \times 3^3 \times 5$ and $1260 = 2^2 \times 3^2 \times 5 \times 7$. Lowest common powers: 2^2 , 3^2 (min of 3^3 , 3^2), 5; the 7 is not common. $HCF = 4 \times 9 \times 5 = 180$. Taking 3^1 gives 60; including 7 and highest powers gives the LCM 3780; dropping a 2 gives 90.

11. (B) $540 = 2^2 \times 3^3 \times 5$, $1260 = 2^2 \times 3^2 \times 5 \times 7$. Highest powers: 2^2 , 3^3 , 5, 7. $LCM = 4 \times 27 \times 5 \times 7 = 3780$. Lowest powers give the HCF 180; raising 5 to 5^2 gives 11340; halving (using $3^2 \times \dots$ wrongly) gives 1890.

12. (D) The common factors of two numbers are exactly the factors of their HCF. $360 = 2^3 \times 3^2 \times 5$, $540 = 2^2 \times 3^3 \times 5$, so $HCF = 2^2 \times 3^2 \times 5 = 180$. Factors of 180 = $(2+1)(2+1)(1+1) = 3 \times 3 \times 2 = 18$. Counting factors of the LCM (1080) gives 24; using exponents directly ($3 \times 3 \times 2$ mis-done) gives smaller wrong counts.

13. (A) $HCF \times LCM = \text{product of the two numbers}$, so $LCM = 7560 \div 6 = 1260$. Multiplying ($7560 \times 6 = 45360$) instead of dividing, or adding 6 (1266), are the slips; 210 ignores the relation.

14. (C) Product of the numbers = $HCF \times LCM = 14 \times 1764 = 24696$. Other = $24696 \div 98 = 252$. Check: $98 = 2 \times 7^2$, $252 = 2^2 \times 3^2 \times 7$; $HCF = 2 \times 7 = 14$ ✓, $LCM = 2^2 \times 3^2 \times 7^2 = 1764$ ✓. Dividing the LCM by 14 ($1764 \div 14 = 126$) instead of the product by 98 is the classic mistake.

15. (A) Product = HCF $\times$ LCM, and product is given as $18 \times \text{HCF}^2$. So $\text{HCF} \times \text{LCM} = 18 \times \text{HCF}^2$, giving $\text{LCM} = 18 \times \text{HCF} = 18 \times 12 = 216$. Using HCF^2 alone (144) or $18 \times \text{HCF}^2$ (2592) misreads the algebra; 18 ignores the HCF.

16. (A) $\text{LCM} = 14 \times 15 = 210$, and product = $\text{HCF} \times \text{LCM} = 15 \times 210 = 3150$. Stopping at the LCM gives 210; squaring the HCF gives 225; an extra factor of 10 gives 31500.

17. (C) 8 and 15 are co-prime, so the numbers are $9 \times 8 = 72$ and $9 \times 15 = 135$. $\text{LCM} = \text{HCF} \times 8 \times 15 = 9 \times 120 = 1080$. Check: $72 = 2^3 \times 3^2$, $135 = 3^3 \times 5$; $\text{LCM} = 2^3 \times 3^3 \times 5 = 1080$ ✓. Forgetting the HCF ($8 \times 15 = 120$), or reading the larger number as the LCM (135), are the traps; multiplying both numbers gives 9720, near 8640.

18. (C) With co-prime ratio 5 : 8, the numbers are $5h$ and $8h$, and $\text{LCM} = 5 \times 8 \times h = 40h$. So $40h = 360 \Rightarrow h = 9$. (The numbers are 45 and 72.) Reading $\text{LCM} \div 8 = 45$ or $\text{LCM} \div 5 = 72$ gives the numbers, not the HCF; 40 forgets to divide.

19. (B) Write $a = 12m$, $b = 12n$ with $\text{HCF}(m, n) = 1$; then $12 \times m \times n = 720$ so $m \times n = 60 = 2^2 \times 3 \times 5$, which has 3 distinct primes. Each prime's whole power goes to m or n , giving $2^3 = 8$ ordered co-prime pairs, hence $8 \div 2 = 4$ unordered pairs: (1,60), (3,20), (4,15), (5,12) $\rightarrow$ (12,720), (36,240), (48,180), (60,144). The rule is $2^{(k-1)}$ with $k = 3$ distinct primes = 4.

20. (D) $a = 15m$, $b = 15n$, $\text{HCF}(m, n) = 1$, so $m \times n = 3150 \div 15 = 210 = 2 \times 3 \times 5 \times 7$ ($k = 4$ distinct primes). Number of unordered co-prime pairs = $2^{(k-1)} = 2^3 = 8$. Using $k = 3$ gives 4; forgetting to halve the 2^4 ordered pairs gives 16.

21. (C) $a = 7m$, $b = 7n$, $\text{HCF}(m, n) = 1$, $m \times n = \text{LCM} \div \text{HCF} = 11 \times 13 \times 17$, which has 3 distinct primes. Unordered co-prime pairs = $2^{(3-1)} = 4$. (The shared 7 is already the HCF and is not a free prime.) Counting the 7 as a fourth free prime wrongly gives 8.

22. (A) Write the numbers as $48a$ and $48b$ with $\text{HCF}(a, b) = 1$; then $48(a + b) = 384$ so $a + b = 8$. Co-prime pairs (a, b) with $a + b = 8$, $a < b$: (1,7) and (3,5) — note (2,6) and (4,4) are not co-prime. So 2 pairs: (48,336) and (144,240). Counting (2,6) as valid gives 3 or 4.

23. (B) For any two numbers, (their product) = $\text{HCF} \times \text{LCM}$, so $n \times 36 = 12 \times 360 = 4320$, giving $n = 4320 / 36 = 120$. Check: $120 = 2^3 \times 3 \times 5$ and $36 = 2^2 \times 3^2$, so $\text{HCF} = 2^2 \times 3 = 12$ and $\text{LCM} = 2^3 \times 3^2 \times 5 = 360$, both correct. Dividing the LCM by the HCF ($360 / 12 = 30$) is not the relationship being tested; $60 = \text{HCF} \times 5$ gives $\text{HCF}(60,36) = 12$ but $\text{LCM}(60,36) = 180$, not 360; 24 and 48 also fail the LCM check.

24. (D) For any valid pair the HCF must divide the LCM. Test: $144 \div 12 = 12$ ✓; $270 \div 9 = 30$ ✓; $300 \div 15 = 20$ ✓; but $204 \div 14 = 14.57...$ is not a whole number, so HCF 14 does not divide LCM 204 — impossible. The refuting check is 'does the HCF divide the LCM exactly?'

25. (C) The number of bouquets must divide all three counts, so it is the HCF. $252 = 2^2 \times 3^2 \times 7$, $378 = 2 \times 3^3 \times 7$, $630 = 2 \times 3^2 \times 5 \times 7$; lowest powers $= 2 \times 3^2 \times 7 = 126$. Taking 3^1 gives 42; missing the 2 gives 63; dropping a 3 and the 7 gives 18.

26. (B) Roses per bouquet $= 252 \div 126 = 2$. (Lilies $378 \div 126 = 3$, tulips $630 \div 126 = 5$.) Dividing the wrong flower count, e.g. $378 \div 126 = 3$ or $630 \div 126 = 5$, gives the lily/tulip counts instead of the roses.

27. (D) Convert: 792 cm and 540 cm. $792 = 2^3 \times 3^2 \times 11$, $540 = 2^2 \times 3^3 \times 5$; HCF (lowest powers) $= 2^2 \times 3^2 = 36$ cm. Taking 3^1 gives 12; missing a 2 gives 18; taking highest powers gives a non-divisor like 108 (which does not divide 792).

28. (B) Tiles along length $= 792 \div 36 = 22$; along width $= 540 \div 36 = 15$. Total $= 22 \times 15 = 330$. Using a wrong tile side or computing $22 + 15$ wrongly, or 12×11 , gives the distractors; 300 comes from rounding the room area carelessly.

29. (A) Convert to cm: 900, 1350, 1890. $900 = 2^2 \times 3^2 \times 5^2$, $1350 = 2 \times 3^3 \times 5^2$, $1890 = 2 \times 3^3 \times 5 \times 7$; lowest powers $= 2 \times 3^2 \times 5 = 90$ cm. Taking $3^1 \times 5$ gives 30; missing a 5 gives 45; taking highest powers gives 270 (not a common divisor).

30. (C) The can size is the HCF. $437 = 19 \times 23$, $551 = 19 \times 29$, $703 = 19 \times 37$; the only common prime factor is 19, so HCF = 19 L. 23, 29 and 37 each divide only one of the volumes, so picking 23 (a factor of 437 alone) or 17 fails on the others.

31. (C) Row size = HCF(1080, 1320). $1080 = 2^3 \times 3^3 \times 5$, $1320 = 2^3 \times 3 \times 5 \times 11$; lowest powers $= 2^3 \times 3 \times 5 = 120$. Taking 3^0 wrongly gives 40-type errors; using 2^2 gives 60; taking the LCM-style highest powers gives 240.

32. (B) Marigold rows $= 1080 \div 120 = 9$; rose rows $= 1320 \div 120 = 11$; total $= 9 + 11 = 20$. Reporting only one kind (9 or 11) or doubling (22) are the slips.

33. (C) String size = HCF. $391 = 17 \times 23$, $425 = 5^2 \times 17$, $493 = 17 \times 29$; the common prime is 17, so HCF = 17. 23 divides only 391, 25 only 425, 29 only 493 — none is common.

34. (C) Plot side = HCF(1517, 1591). $1591 - 1517 = 74 = 2 \times 37$; $1517 = 37 \times 41$, $1591 = 37 \times 43$, so HCF = 37 m. Plots along 1591 m $= 1591 \div 37 = 43$. Computing along the other side ($1517 \div 37 = 41$) gives 41; mistaking 37 for the count gives 37.

35. (A) Bundle size = HCF. $1224 = 2^3 \times 3^2 \times 17$, $1632 = 2^5 \times 3 \times 17$, $2040 = 2^3 \times 3 \times 5 \times 17$; lowest powers $= 2^3 \times 3 \times 17 = 408$. Taking 2^2 gives 204; dropping the 3 gives 136; taking highest powers gives a non-divisor like 612.

36. (D) Bag size = HCF(84, 126, 210). $84 = 2^2 \times 3 \times 7$, $126 = 2 \times 3^2 \times 7$, $210 = 2 \times 3 \times 5 \times 7$; lowest powers $= 2 \times 3 \times 7 = 42$ kg. Bags $= 84/42 + 126/42 + 210/42 = 2 + 3 + 5 = 10$. Reporting the bag size (42) instead of the count, or doubling the 10, are the traps.

- 37. (A)** Next together = LCM(18, 24, 32). $18 = 2 \times 3^2$, $24 = 2^3 \times 3$, $32 = 2^5$; highest powers = $2^5 \times 3^2 = 32 \times 9 = 288$. Using 2^4 gives 144; doubling gives 576; missing a 3 gives 96.
- 38. (D)** LCM(12, 18, 21, 28): $12 = 2^2 \times 3$, $18 = 2 \times 3^2$, $21 = 3 \times 7$, $28 = 2^2 \times 7$; highest powers = $2^2 \times 3^2 \times 7 = 4 \times 9 \times 7 = 252$. Doubling gives 504; taking 3^1 gives 84; using $2 \times 3^2 \times 7$ gives 126.
- 39. (A)** LCM(12, 21, 28) = $2^2 \times 3 \times 7 = 84$ minutes = 1 h 24 min. Adding to 9:00 am gives 10:24 am. Doubling to 168 min gives 11:48 am; 9:84 am is the impossible 'add minutes without carrying'; 10:00 (60 min) ignores the 84.
- 40. (D)** They meet again after LCM(56, 70, 84). $56 = 2^3 \times 7$, $70 = 2 \times 5 \times 7$, $84 = 2^2 \times 3 \times 7$; highest powers = $2^3 \times 3 \times 5 \times 7 = 840$ s. Fastest runner has the smallest lap time (56 s), so laps = $840 \div 56 = 15$. Using 70 s gives 12; using 84 s gives 10.
- 41. (B)** They meet after LCM(54, 72, 90). $54 = 2 \times 3^3$, $72 = 2^3 \times 3^2$, $90 = 2 \times 3^2 \times 5$; highest powers = $2^3 \times 3^3 \times 5 = 8 \times 27 \times 5 = 1080$ s. The slowest cyclist has the LONGEST lap time (90 s), so laps = $1080 \div 90 = 12$. Using 72 s gives 15; using 54 s gives 20.
- 42. (A)** Alignment returns after a whole number of turns of each gear, i.e. after LCM(36, 48) teeth pass any meshing point. $36 = 2^2 \times 3^2$, $48 = 2^4 \times 3$; LCM = $2^4 \times 3^2 = 144$. Since teeth pass at the same rate at the mesh, the smaller gear has 144 teeth pass (= 4 full turns). Using HCF (12), doubling (288), or multiplying 36×48 (1728) are the traps.
- 43. (A)** Together every LCM(9, 12, 15) = 180 s = 3 min. In 30 minutes there are $30 \div 3 = 10$ such moments after the start. Counting the start too gives 11; halving gives 5.
- 44. (B)** LCM(6, 8, 12, 16): $6 = 2 \times 3$, $8 = 2^3$, $12 = 2^2 \times 3$, $16 = 2^4$; highest powers = $2^4 \times 3 = 48$ s. In 240 s the joint tolls are at 0, 48, 96, 144, 192, 240 $\rightarrow$ 6 tolls. Excluding the start gives 5; using 24 s gives 10.
- 45. (C)** LCM(36, 48) = $2^4 \times 3^2 = 144$ min = 2 h 24 min. Adding to 7:00 am gives 9:24 am. Using 84 min (a wrong LCM) gives 8:24 am; doubling 144 gives 11:48 am.
- 46. (A)** LCM(48, 72, 108): $48 = 2^4 \times 3$, $72 = 2^3 \times 3^2$, $108 = 2^2 \times 3^3$; highest powers = $2^4 \times 3^3 = 16 \times 27 = 432$. Using 3^2 gives 144; using 2^3 gives 216; doubling gives 864.
- 47. (B)** Such a number is LCM(15, 20, 35) $\times k + 8$. LCM = $2^2 \times 3 \times 5 \times 7 = 420$; smallest ($k = 1$) = $420 + 8 = 428$. Forgetting the +8 gives 420; subtracting 8 gives 412; doubling the LCM gives 848.
- 48. (B)** For a common UNKNOWN remainder, take the HCF of the pairwise differences. $4665 - 1305 = 3360$, $6905 - 4665 = 2240$, $6905 - 1305 = 5600$. HCF(3360, 2240, 5600): $3360 = 2^5 \times 3 \times 5 \times 7$, $2240 = 2^6 \times 5 \times 7$, $5600 = 2^5 \times 5^2 \times 7$; lowest powers = $2^5 \times 5 \times 7 = 1120$. Using 2^4 gives 560; using 2^7 gives 2240; dropping powers gives 140.

49. (D) The number is $\text{LCM}(8, 12, 18) \times k + 5$. $\text{LCM} = 2^3 \times 3^2 = 72$; smallest above 5 takes $k = 1$: $72 + 5 = 77$. Forgetting the +5 gives 72; subtracting gives 67; taking $k = 2$ gives 149.

50. (A) $\text{LCM}(8, 12, 20) = 2^3 \times 3 \times 5 = 120$. Greatest 3-digit multiple of 120 is 960 (120×8). Add 7: $960 + 7 = 967 (\leq 999 \checkmark)$. The next multiple $1080 + 7$ exceeds 3 digits. Forgetting +7 gives 960; using LCM 840 gives 847; 975 is $960 + 15$, a miscount.

51. (D) We need $N + 11 = \text{LCM}(18, 24, 30) \times k$. $\text{LCM} = 2^3 \times 3^2 \times 5 = 360$; smallest positive N takes $k = 1$: $N = 360 - 11 = 349$. Adding instead of subtracting gives 371; taking the LCM itself gives 360; $k = 2$ gives 709.

52. (C) Numbers leaving remainder 1 on 2–6 are $\text{LCM}(2, \dots, 6) \cdot k + 1 = 60k + 1$: 61, 121, 181, 241, 301, ... Test divisibility by 7: 61(no), 121(no), 181(no), 241(no), $301 = 7 \times 43 \checkmark$. So 301. Stopping at 61 ignores the $\div 7$ condition; 421 is the next such value.

53. (D) In both cases the number falls 5 short of a multiple: $12 - 7 = 5$ and $18 - 13 = 5$. So $N + 5$ is divisible by both 12 and 18, i.e. by $\text{LCM}(12, 18) = 36$. Smallest $N = 36 - 5 = 31$. Check: $31 \div 12 \rightarrow r 7 \checkmark$, $31 \div 18 \rightarrow r 13 \checkmark$. Adding 5 gives 41/43-type errors; 36 forgets the shift.

54. (B) $9 = 3^2$, $16 = 2^4$, $35 = 5 \times 7$ share no common prime across any pair, so all three pairs have HCF 1 $\checkmark$. In '6, 25, 14', 6 and 14 share 2. In '8, 9, 21', 9 and 21 share 3. In '15, 22, 33', 15 and 33 share 3 (and 22, 33 share 11).

55. (B) A common multiple of 16 and 36 is a multiple of their LCM. $16 = 2^4$, $36 = 2^2 \times 3^2$; $\text{LCM} = 2^4 \times 3^2 = 144$. The smallest such number is 144 itself. Multiplying $16 \times 36 = 576$ over-counts; the HCF 4 confuses divisor with multiple; 288 is 2×144 .

56. (A) If a and b share no prime factor, then neither do their cubes, so $\text{HCF}(a^3, b^3) = 1$. Raising co-prime numbers to any power keeps them co-prime. The other options assume a shared factor that does not exist.

57. (A) Co-prime factors above 1 mean two distinct prime factors. $1517 = 37 \times 41$ (both prime, so co-prime) $\checkmark$. '1 and 1517' violates 'each greater than 1'; $23 \times 66 = 1518 \neq 1517$; $31 \times 49 = 1519 \neq 1517$.

58. (A) 'Always divisible' means by the LCM of the stated divisors. $\text{LCM}(9, 10) = 2 \times 3^2 \times 5 = 90$. We may not assume an extra factor of 2 (which 180 or 360 would require). 45 misses the factor 2 from 10.

59. (D) Common multiples of 12 and 15 are multiples of $\text{LCM}(12, 15) = 60$. Multiples of 60 up to 500: 60, 120, ..., 480 $\rightarrow 480 \div 60 = 8$. Using 50 ($\approx 500/12$ wrongly) or counting multiples of 15 alone gives wrong counts; 4 halves the answer.

60. (B) Convert: 425, 680, 1020 cm. $425 = 5^2 \times 17$, $680 = 2^3 \times 5 \times 17$, $1020 = 2^2 \times 3 \times 5 \times 17$; lowest common powers = $5 \times 17 = 85$ cm. Taking only 17 or only 5 drops a common factor; $170 = 2 \times 85$ is not a common divisor (2 does not divide 425).